MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 10: Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In a circle of radius 7 cm, tangent PT is drawn from external point P. If $OT = 7$ cm and $PT = 24$ cm, then the distance OP is: [1]

(A) 25 cm (B) 26 cm (C) 31 cm (D) $\sqrt{527}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 25 cm

Justification: Since $OT \perp PT$, in right $\triangle OTP$: $OP = \sqrt{OT^2 + PT^2} = \sqrt{7^2 + 24^2} = \sqrt{625} =$ **25** cm.

Q2. If two tangents inclined at an angle of 60° are drawn to a circle of radius 3 cm, then the length of each tangent is: [1]

(A) $\frac{3}{2}\sqrt{3}$ cm (B) 6 cm (C) 3 cm (D) $3\sqrt{3}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $3\sqrt{3}$ cm

Justification: Line OP bisects the 60° angle into 30°. In right $\triangle OAP$: $\tan 30° = \frac{3}{PA} \implies PA =$ **$3\sqrt{3}$** cm.

Q3. If a parallelogram circumscribes a circle, then the parallelogram must be a: [1]

(A) Rectangle (B) Square (C) Rhombus (D) Trapezium

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) Rhombus

Justification: From Pitot's theorem ($AB + CD = AD + BC$) and parallelogram opposite sides ($AB = CD, AD = BC$), $2AB = 2BC \implies AB = BC$. Thus all sides are equal, so it is a **Rhombus**.

Q4. Two concentric circles have radii a and b ($a > b$). A chord of length $2c$ of the larger circle touches the smaller circle. The relation between a, b, c is: [1]

(A) $a^2 = b^2 + c^2$ (B) $b^2 = a^2 + c^2$ (C) $c^2 = a^2 + b^2$ (D) $a + b = 2c$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $a^2 = b^2 + c^2$

Justification: The radius b to the point of contact bisects the chord of length $2c$ perpendicularly into segments of length c. By Pythagoras theorem: $\boxed{\mathbf{a^2 = b^2 + c^2}}$.

Q5. **Assertion (A):** At most two tangents can be drawn to a circle from a point lying outside the circle. **Reason (R):** From a point lying strictly inside the circle, exactly one tangent can be drawn to the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

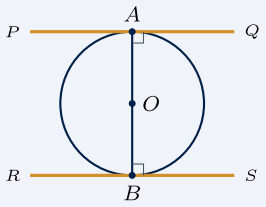
Justification: Assertion (A) is true (Theorem 10.2: exactly two tangents from an exterior point). Reason (R) is false because every line through an interior point intersects the circle at two distinct points (always a secant), so 0 tangents can be drawn.

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Prove that the tangents drawn at the end points of a diameter of a circle are parallel. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Diameter AB of circle centred at O; tangents PQ at A and RS at B.



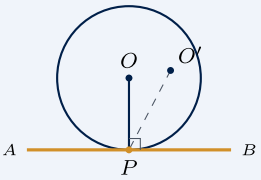
Step 1: Radius–Tangent Angles [1 Mark]
By Theorem 10.1, $OA \perp PQ \implies \angle OAQ = 90°$ and $OB \perp RS \implies \angle OBS = 90°$ (similarly $\angle OAP = 90°, \angle OBR = 90°$).

Step 2: Alternate Interior Angles [1 Mark]
$\angle PAB = \angle ABS = 90°$ (equal alternate interior angles made by transversal AB).

$\boxed{\therefore PQ \parallel RS \quad \text{[Proved]}}$

Q7. Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Tangent AB touches circle centred at O at point of contact P.

To Prove: The perpendicular to AB at P passes through O. [1 Mark]

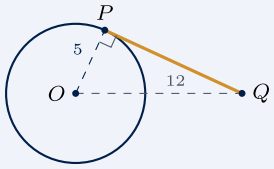
Proof by Contradiction: [2 Marks]

1. Suppose the perpendicular to AB at P does not pass through O, and passes through another point O'. Thus $\angle O'PB = 90°$.
2. Join OP. By Theorem 10.1 (radius $\perp$ tangent), $\angle OPB = 90°$.
3. Thus $\angle O'PB = \angle OPB = 90°$, which is only possible if rays PO' and PO coincide.

$\therefore$ The perpendicular at P passes through O [Proved]

Q8. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. Find the length PQ and the area of $\triangle OPQ$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Finding Length of Tangent PQ [1.5 Marks]

Since $OP \perp PQ$ at P (Theorem 10.1), $\triangle OPQ$ is right-angled at P:

$$PQ = \sqrt{OQ^2 - OP^2} = \sqrt{12^2 - 5^2} = \sqrt{144 - 25} = \sqrt{119} \text{ cm}$$

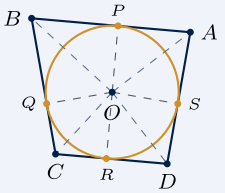
Step 2: Area of $\triangle OPQ$ [1.5 Marks]

$$\text{Area}(\triangle OPQ) = \frac{1}{2} \times OP \times PQ = \frac{1}{2} \times 5 \times \sqrt{119} = \frac{5\sqrt{119}}{2} \text{ cm}^2$$

$$\boxed{PQ = \sqrt{\mathbf{119}} \text{ cm}, \quad \text{Area}(\triangle OPQ) = \frac{\mathbf{5\sqrt{119}}}{\mathbf{2}} \text{ cm}^{\mathbf{2}} \ (\approx \mathbf{27.27} \text{ cm}^{\mathbf{2}})}$$

Q9. In a quadrilateral $ABCD$ circumscribing a circle, prove that $\angle AOB + \angle COD = 180°$ and $\angle BOC + \angle AOD = 180°$, where O is the centre of the circle. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Let the circle touch AB, BC, CD, DA at P, Q, R, S. Join OA, OB, OC, OD and radii OP, OQ, OR, OS. [0.5 Mark]

Proof & Deduction: [2.5 Marks]

1. By SSS congruence, $\triangle OAP \cong \triangle OAS \implies \angle AOP = \angle AOS$. Similarly, $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$, and $\angle DOR = \angle DOS$.
2. Total angle around O: $2(\angle AOP + \angle BOP + \angle COR + \angle DOR) = 360° \implies \angle AOB + \angle COD = 180°$.
3. Grouping adjacent pairs $2(\angle BOQ + \angle COQ + \angle DOS + \angle AOS) = 360° \implies \angle BOC + \angle AOD = 180°$.

$\angle AOB + \angle COD = 180°$ and $\angle BOC + \angle AOD = 180°$ [Proved]

Q10. Municipal Triangular Park with Inscribed Circular Fountain

A municipal corporation plans an ornamental circular fountain inscribed within a triangular park ABC. The park boundaries measure $AB = 15$ m, $BC = 14$ m, $AC = 13$ m. The fountain touches boundary BC at point D, AC at E, and AB at F. *(Note: All calculations strictly follow the problem statement units in metres, m.)*

(i) Calculate the semi-perimeter s of the triangular park. [1]

(ii) Find the lengths of tangent segments BD and CD on side BC. [1]

(iii) Calculate the radius r of the circular fountain if the area of $\triangle ABC = 84$ m^2, OR find the length of tangent segment AF. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Semi-Perimeter s: [1 Mark]

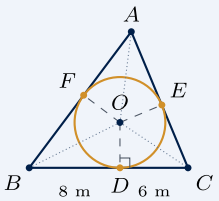
$s = \frac{AB+BC+CA}{2} = \frac{15+14+13}{2} = \frac{42}{2} = \boxed{\mathbf{21}\text{ m}}$.

(ii) Tangent Segments BD and CD: [1 Mark]

Let $AF = AE = x$, $BD = BF = y$, $CD = CE = z$.

Since $y + z = 14$ and $x + y + z = 21 \implies x = 7$ m.

Thus $BD = y = 15 - 7 = \boxed{\mathbf{8}\text{ m}}$, $CD = z = 13 - 7 = \boxed{\mathbf{6}\text{ m}}$.



(iii) Main Choice — Radius r of Circular Fountain: [2 Marks]

Joining OA, OB, OC: Area$(\triangle ABC)$ = Area$(\triangle OBC)$ + Area$(\triangle OCA)$ + Area$(\triangle OAB) = \frac{1}{2}r(BC + AC + AB) = r \cdot s$.

Given Area $= 84$ m^2: $84 = r \times 21 \implies \boxed{r = \mathbf{4}\text{ m}}$.

— OR — (Alternative Choice — Length of Tangent Segment AF): [2 Marks]

Using $AF = s - BC = 21 - 14$ (or from $x + 14 = 21$), we get $\boxed{AF = \mathbf{7}\text{ m}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★